

ASCII MASTER FLOW – PANEL 1/12: Ocean-current force chain

WIND STRESS -> surface momentum

CORIOLIS -> turning by hemisphere

PRESSURE GRADIENT -> geostrophic tendency

BASIN + FRICTION -> realised current path

MUST REMEMBER: Ocean structure links temperature, salinity and density to stratification and circulation; wind stress, Coriolis, Ekman transport, gyres, upwelling, thermohaline overturning, tides and Indian Ocean monsoon reversal culminate in IOD-ENSO interactions.

MUST REMEMBER: Ocean structure links temperature, salinity and density to stratification and circulation; wind stress, Coriolis, Ekman transport, gyres, upwelling, thermohaline overturning, tides and Indian Ocean monsoon reversal culminate in IOD-ENSO interactions.

ASCII MASTER FLOW – PANEL 2/12: Density-overturning chain

COOLING / EVAPORATION -> density increase

FRESHWATER / WARMING -> density decrease

DENSE WATER -> sinking and spreading

MIXING + TOPOGRAPHY -> overturning pathways

ASCII MASTER FLOW – PANEL 3/12: Gyre comparison

NORTH SUBTROPICS -> clockwise tendency

SOUTH SUBTROPICS -> anticlockwise tendency

WESTERN LIMB -> narrow, swift, warm

EASTERN LIMB -> broad, cool, upwelling-prone

ASCII MASTER FLOW – PANEL 4/12: Upwelling-food web

ALONGSHORE WIND -> Ekman transport offshore

DEEPER WATER RISES -> nutrient supply

PHYTOPLANKTON -> zooplankton -> fish

LIMITS -> oxygen, light, habitat, fishing pressure

ASCII MASTER FLOW – PANEL 5/12: Tide geometry

NEW / FULL MOON -> aligned forces

SPRING TIDE -> larger range

QUARTER MOON -> partial offset

NEAP TIDE -> smaller range

ASCII MASTER FLOW – PANEL 6/12: Local tidal range

ASTRONOMICAL FORCING -> incoming tide
SHELF DEPTH + BAY SHAPE -> amplification
RESONANCE + FRICTION -> timing and height
RULE -> same forcing, different local range

ASCII MASTER FLOW – PANEL 7/12: Salinity budget

EVAPORATION + BRINE -> salinity rises

RAIN + RIVER + MELT -> salinity falls

ADVECTION + MIXING -> redistribute salt

TEMP + SALINITY -> density and stratification

ASCII MASTER FLOW – PANEL 8/12: Arabian-Bay contrast

ARABIAN SEA -> stronger evaporation

BAY OF BENGAL -> rain + large river input

GENERAL RESULT -> Arabian Sea more saline

QUALIFIER -> season and local mixing matter

ASCII MASTER FLOW – PANEL 9/12: Monsoon-current switch

SW MONSOON -> winds drive summer reorganisation

SOMALI SECTOR -> strong current + upwelling

NE MONSOON -> northern flows reverse/reorganise

MAP RULE -> label every arrow by season

ASCII MASTER FLOW – PANEL 10/12: IOD coupled seesaw

WEST-EAST SST CONTRAST -> pressure gradient

WINDS -> thermocline and upwelling response

CONVECTION -> rainfall shifts zonally

INDIA EFFECT -> conditional on timing + ENSO + MJO

CLOSE DISTINCTION: Wave transmits energy, current transports water and tide is astronomical sea-level oscillation; spring/neap are alignment effects, not seasons, and positive IOD is a west-minus-east Indian Ocean SST gradient rather than simply a warm western ocean.

CLOSE DISTINCTION: Wave transmits energy, current transports water and tide is astronomical sea-level oscillation; spring/neap are alignment effects, not seasons, and positive IOD is a west-minus-east Indian Ocean SST gradient rather than simply a warm western ocean.

ASCII MASTER FLOW – PANEL 11/12: Observation-to-service chain

OMNI / RAMA -> ocean + weather observations

INCOIS -> data management and products

MODELS -> analysis and forecast ensembles

USER -> marine service with uncertainty

EVIDENCE LIMIT: IOD and ENSO modulate probabilities rather than determine Indian rainfall;
event phase, season, interaction, basin background and model/source date must qualify
fisheries, cyclone or monsoon conclusions.

EVIDENCE LIMIT: IOD and ENSO modulate probabilities rather than determine Indian rainfall;
event phase, season, interaction, basin background and model/source date must qualify
fisheries, cyclone or monsoon conclusions.

ASCII MASTER FLOW – PANEL 12/12: Ocean answer spine

DEFINE -> depth, scale and season

DERIVE -> force, density or gravity

MAP -> basin, current, tide or salinity

QUALIFY -> variability, ecology and evidence date